SYSC 4907A – Project Proposal

Group 76: AI Study Buddy

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Objectives

The **Study Buddy** project aims to design and implement an AI-powered focus and learning assistant that helps students improve concentration, organization, and comprehension during study sessions. The system combines a hardware device (for focus monitoring and voice interaction) with a web platform (for course organization and AI-assisted studying) to provide a complete learning support ecosystem.

The device: uses a camera to monitor focus, provides immediate feedback through LED, vibration, or sound when distraction is detected, and allows voice interaction so the student can speak to the Study Buddy and hear it respond verbally. On the software side, students can also type questions in a chat interface and receive text-based answers generated by an AI model that references their uploaded study materials (notes, slides, or textbooks).

The web platform: enables session organization by course, document uploads, flashcard generation, summaries, and interactive question-answering, all grounded in the student's own materials.

1. **Enhance Student Focus:**
   Create a hardware-assisted attention monitoring system that detects distraction through computer vision and provides immediate feedback using visual, auditory, or haptic signals.

2. **Enable Voice and Text Interaction:**
   Develop a dual-interface system where students can speak directly to the device or chat via the web platform. The AI will respond verbally through the device's speaker or textually through chat, based on the mode of interaction.

3. **Organize and Track Study Sessions:**
   Provide a structured web interface for managing courses, study sessions, and documents while tracking focus metrics and productivity over time.

4. **Facilitate Deeper Comprehension via AI Assistance:**
   Integrate an LLM-based assistant that can answer questions, generate flashcards, and create summaries from uploaded documents, helping students understand and retain material more effectively.

Functional and Non-Functional Requirements

Functional Requirements	Measurement Criteria
1. Detect distraction (e.g., user looking away or using phone) via camera.	≥80% detection accuracy, <10% false positives across 20 test sessions.
2. Trigger LED, sound, or vibration feedback when distraction lasts >10s.	Reaction time <1s after detection event.
3. Allow voice commands and questions to the Study Buddy device.	95% ASR transcription accuracy on clear speech; latency ≤4s per query.
4. Respond verbally to user queries via on-device speaker.	Text-to-speech latency ≤2s; speech output intelligibility score ≥85%.
5. Allow typed questions through the web platform with text-based responses.	Text responses returned within ≤4s for short queries.
6. Enable course and document management (upload, organize, delete).	CRUD operations completed within ≤2s latency per action.
7. Generate flashcards and summaries from uploaded materials.	≥20 flashcards generated per 50-page document in <60s.
8. Record and visualize session focus history and metrics.	Focus graph rendered within >30s after session completion.

Non-Functional Requirement	Measurement Criteria
Usability: The system must be intuitive and easy to operate for new users.	System Usability Scale (SUS) ≥80 from ≥5 test users.
Reliability: Device must function offline and sync data upon reconnection.	≥95% successful event replay rate after connection loss.
Scalability: Backend must handle multiple users concurrently.	50 concurrent sessions with average API latency <2s.
Privacy: No raw audio or video stored or transmitted externally.	Manual code audit confirms only processed metadata is retained.

Maintainability: Modular codebase with testing and CI integration.	≥70% code coverage; all modules pass integration tests.
Performance: LLM responses and flashcard generation must be timely.	Mean total response latency ≤5s for standard queries.
Availability: System must remain accessible during live demonstrations.	≥95% uptime during testing and demo phase.

Progress Measurement Plan / Proposed Timetable:

Phase	Timeline	Key Deliverables	Progress Measurement
Phase 1: Design & Setup	**Oct 17-Nov17**	• Final system architecture and diagrams • Hardware components assembled and tested • Backend and web app scaffolding created	- Design documents approved by team - Hardware test (camera, LED, vibration) passes basic functionality - Code repository and CI/CD pipeline active
Phase 2: Focus Monitoring MVP	**Nov18 – Dec 18**	• **Progress Report** • Focus detection model (face/gaze) • LED/sound/vibration feedback loop • Live focus state visualization on web	- ≥ 80 % detection accuracy in controlled tests - Feedback triggered in < 1 s - Progress report and short demo video submitted by Dec 5
Phase 3: AI & Document Processing	**Dec 19 – Jan 19**	• Document upload and embedding pipeline • Retrieval-augmented Q&A prototype • Citation and flashcard generation	- Successful end-to-end question answering from uploaded PDFs - ≥ 85 % citation accuracy - Flashcards generated in < 60 s
Phase 4: Voice & Chat Integration	**Jan 20 - Feb 20**	• **Oral Presentation** • Speech-to-text and text-to-speech working on device • Web chat interface connected to backend • Real-time dual-mode interaction	- Voice latency ≤ 4 s E2E - 95 % ASR accuracy in quiet environment - Oral presentation demo (Jan 19–23) completed
Phase 5: Integration & User Testing	**Feb 21 - Mar 19**	• Full hardware-software integration • Usability testing with sample users	- ≥ 90 % functional tests passed - SUS ≥ 80 from user survey - Offline sync validated (≥ 95 % replay success)

		• Focus and learning analytics dashboard	
Phase 6: Finalization & Reporting	**Mar 20 - April 8**	**• Final system report and poster** **• Public demo of final prototype** **• Documentation and evaluation results**	- All objectives verified against metrics - Final report submitted Mar 20 - Successful live demo Mar 27

Background

Current software solutions for AI-involved learning and studying revolve around study tools like quiz generation, summarization, grammar checking, flashcard generation, and other tools of that nature. Services like Penseum, Mindgrasp, and Studypotion have focused on these tools and have done an incredible job at tailoring either to specific subjects or specific learning styles. Pearson study prep is one example of tailoring to a specific learning style (video learning) by suggesting videos on specific subjects based on the curriculum and reference text given to it.

On the hardware aspect, companies like *Miko* and *Misa Robotics* have created products focused on child education or personal assistance. *Miko* incorporates learning into games and answering questions a curious child might ask, such as "Why does the sky change colour in the evening?" and reminder and scheduling capabilities.

These solutions either focus on performing a task that a focused learner would use or bringing in a social aspect to personal assistance technology, as in the case of *Misa Robotics*. There is also a significant focus on children's learning in the design of the current commercial robots involving AI. The project takes a different approach by merging features from both software and hardware and including a distraction monitoring feature. The goal is to serve as an accountability and learning companion for all individuals studying alone and/or who are more susceptible to distractions.

Description

The team plans to create a system that allows users to chat with an AI assistant on specific topics and concepts in reference to a textbook or other text that has been given to the assistant previously. The systems would provide typical AI study tools: quiz and flashcard generation, summarization, e.t.c. It will focus on text-based study tools. The system would also include a monitoring subsystem that observes the user for distraction. If the user is distracted, it produces auditory and visual cues as a reminder to focus.

Relation to Degrees

Student A: Software Engineering (AI & Backend Integration)

This student's primary role will focus on designing and implementing the backend architecture and AI integration pipeline, which includes:

- Developing the FastAPI service layer and database schema (PostgreSQL with pgvector).
- Implementing retrieval-augmented generation (RAG) workflows for the LLM-based Q&A and flashcard generation system.
- Handling authentication, API endpoints, and system integration with the frontend and hardware components.

These responsibilities directly relate to the Software Engineering curriculum, particularly in areas of:

- Software architecture design (SYSC 3120, SYSC 4120),
- Databases and distributed systems (COMP 3005),
- AI application integration, and
- Testing and documentation processes as required by professional software engineering standards (SYSC 4101).

This student's contribution demonstrates the full software development life cycle (SDLC) from requirements analysis and system design to deployment and testing as expected from a graduating Software Engineer.

Student B: Software Engineering (Frontend & User Experience)

This student will focus on building the web platform interface and ensuring an effective user experience.

Key responsibilities include:

- Designing and implementing the Next.js frontend application.
- Developing real-time dashboards to visualize focus metrics and session data.
- Creating the chat interface and managing the voice–text interaction integration.
- Conducting usability and accessibility testing to meet design best practices.

These tasks align directly with the Software Engineering program's emphasis on:

- Human–Computer Interaction (COMP 3008, SYSC 4906),
- Software design patterns and modularity (SYSC 4120, SYSC 3110),
- Client–server communication, and
- Performance evaluation and optimization (SYSC 4101, SYSC 4806, SYSC 3303).

The student's work ensures that the system is not only functional but also intuitive, maintainable, and user-centred, reflecting key professional software design principles.

Student C: Software Engineering (AI Evaluation & System Integration)
This student will lead the AI model evaluation and system-level integration testing:

- Evaluating the accuracy, latency, and reliability of the AI's responses.
- Implementing and validating data flow between the backend, frontend, and hardware.
- Developing automated testing suites, CI/CD pipelines, and system performance metrics.
- Conducting user testing to assess learning improvement and usability (SUS evaluation).

This work corresponds closely with the Software Engineering program outcomes related to:

- Quality assurance and testing methodologies (SYSC 4101),
- Software metrics and verification (SYSC 4101, SYSC 4806),
- Systems integration, and
- Project management and documentation (SYSC 4120, SYSC 4806)

Through these responsibilities, this student applies principles of engineering validation and process improvement, ensuring that the product is both reliable and maintainable.

Student D: Computer Systems Engineering (Hardware, Embedded & CV Systems)
This student's primary responsibility lies in the hardware and embedded systems domain, bridging the physical and digital components of the Study Buddy:

- Setting up the Raspberry Pi device and integrating peripherals (camera, microphone, LED, vibration motor, speaker).
- Implementing real-time computer vision algorithms (OpenCV, MediaPipe) to detect user focus and distraction.
- Managing sensor input/output control, signal processing, and on-device privacy enforcement.
- Ensuring reliable communication between the device and backend via WebSocket or message queuing telemetry transport (MQTT) protocols.

These tasks directly apply core competencies of the Computer Systems Engineering program, including:

- Embedded system design and interfacing (SYSC 3310, SYSC 3320)
- Digital signal processing (SYSC 3501)
- Hardware–software integration, and (SYSC 3310, SYSC 4805)
- Real-time systems development. (SYSC 3303)

This role highlights the practical application of hardware design and embedded software, demonstrating how system-level engineering principles are used to create a responsive, privacy-preserving AI device.

Group's Skills in Accomplishing the Project

Software Engineering Expertise:
Three team members are final-year Software Engineering students, each bringing a strong foundation in software architecture, full-stack web development, AI integration, systems testing, and user experience design.

- These members are responsible for designing both backend and frontend systems, including developing FastAPI services, creating the Next.js web platform, and integrating AI-powered features such as retrieval-augmented generation for question-answering and flashcard generation.
- Their coursework enables them to apply advanced concepts in databases, distributed systems, human-computer interaction, client-server models, and quality assurance, ensuring robust, scalable, and maintainable code.
- They also lead usability testing, CI/CD pipeline management, and system-wide integration and evaluation to guarantee a user-friendly and reliable product.

Computer Systems Engineering Expertise:
One team member specializes in Computer Systems Engineering, focusing on embedded hardware, real-time systems, and hardware-software integration.

- This member leads the design and implementation of the hardware platform, from setting up the Raspberry Pi system and peripherals (camera, LEDs, microphones, etc.) to developing real-time computer vision algorithms and managing sensor-actuator interfaces.
- They ensure efficient, privacy-preserving, and robust communication between the hardware and backend, leveraging experience in embedded systems, digital signal processing, and mechatronics.

Integrated Skills and Collaboration:
Together, the team covers the complete development lifecycle:

- Software architecture and modularity for maintainable code.
- Human-centered design and accessibility best practices.
- Cutting-edge AI evaluation and automation for the learning assistant pipeline.
- Hardware interfacing, signal processing, and embedded privacy for the focus-monitoring device.
- Agile project management and test-driven methodologies, enabling iterative refinement, risk mitigation, and adaptability to emerging project needs.
- Ethical, privacy, and legal compliance, ensuring adherence to data protection laws and ethical standards in user monitoring.

Methods Used and Relation to Degree

- **Software project management**
  - Agile development: The project has aspects for which the team does not have extensive experience with such as the AI features. With that fact in place, it is reasonable to expect that there will be requirements that are either overestimated or underestimated. The team will use the iterative and adaptive nature of the Agile approach to software management to handle these expectations and ensure that the features created are built to the best extent possible.
- **Software architecture**
  - Object-oriented design: Following this design principle will allow for easier separation of concerns and fewer complications when modifying or replacing features. It also allows for easier designation of tasks between members of the system.
  - Pipeline architecture: A pipeline design will be used to manage the AI interactions within the system. For the AI study tools, requests will follow a process of processing and filtering for the tools to perform their tasks. The AI detection of distraction will take the video images and feed them through the model to determine the next actions the system would take.
  - Domain partitioning of components: The system will be composed of modules that handle the different aspects of the system, such as user input, communication with external services, and the operation of output devices.
  - Model-View-Controller pattern: This will be used to manage the Graphical User Interface (GUI) needed for easier interaction with the user. The pattern will also allow for an easier time changing visual elements without having to deeply change the rest of the system.
- **Testing**
  - Test Driven Development (TDD): Test Driven Development follows the philosophy of creating tests before creating code. The tests serve as a check to build confidence that the code being written works the way it should work. It has proven to reduce defects (accidental bugs created in code by developers) and increase the maintainability of the systems where the philosophy is practiced (Makinen & Münch, 2014, 13). The team will apply the philosophy to the project for those advantages.
  - Unit testing: Unit testing is a practice under TDD that concerns itself with testing the smallest aspects of a program that can be isolated, such as functions, modules, and classes. It follows the rule that all tests are repeatable and isolated, i.e. they do not depend on any other tests. The practice helps ensure confidence that the individual units of the system work as expected. The team will apply this practice

to the project to build confidence that changes made to the system units do not affect the expected behaviour of those system units.

  - Integration testing: Integration testing exercises the most important interactions of the system for correctness. In the case of the project, that would be the communications with the necessary external services and the interface between the system and its hardware components.

- **Mechatronics**
  - Software engineering deals with processes involved in developing and maintaining software in all aspects. This includes the software that handles hardware control. Courses such as ECOR 1044 (2022) on mechatronics and SYSC 3310 (2024) on real-time systems have provided experience with the interaction of software with hardware components. This experience will be essential when interfacing with the camera, LEDs, etc., that will be used in the project.

Project Risks and Mitigation Strategies

Risk	Concerns	Mitigation Strategies
Privacy and Data Protection Breach: Cameras and AI systems would collect sensitive behavioural data (e.g., facial expressions, gaze direction, body language)	• Non-subject individuals are being recorded (i.e. people in the background). • Misuse/leakage of video data. • Recording the participant without proper consent.	• Explicit informed consent is required from all participants. • Data must be processed locally and anonymized. • The project must comply with regulations and local privacy laws.
Ethical and Trust Issues: The project could be seen as surveillance or invasive, potentially harming trust between students and the group.	• Students feel watched and change their behaviour unnaturally. • Perception that the system is unjustly ranking focus levels.	• Maintain transparency about what's being recorded and why. • Focus on voluntary participation and educational benefits over monitoring.
A.I. Bias and Inaccuracy: The model may misinterpret behaviours depending on factors like ethnicity, disability, lighting, or camera quality	• AI misclassifies cultural or neurodivergent expressions as "disengaged." • Students with different learning styles are unfairly labelled as unfocused.	• Train the model on diverse, representative datasets. • Continuous evaluation of the system's bias and performance. • Involve human evaluators to interpret AI results, avoiding sole dependence on automation.
Technical Difficulties: Hardware or software failures could lead to incorrect feedback or potential data loss.	• Cameras malfunction or produce poor-quality footage. • AI models produce inconsistent/ misleading results.	• Implement Backup systems/servers to save the system's current state. • The model and sensors go through

		recurrent testing and calibration. ● Implement error-handling and data validation protocols.
Social and Psychological Risk: Reduced natural behaviour, stress, or burnout	● Students may feel anxious or self-conscious about being constantly monitored	● Frame the system as a personal improvement tool ● Include opt-out options and mental health considerations
Project Management Risk: Misalignment between technical goals, ethics and educational outcomes	● Overpromising results ● Poor system integration	● Set realistic goals and milestones ● Engage stakeholders (students, coordinators) early. ● Conduct agile testing during development stages.

List of Special Components and Facilities Required

<u>Hardware</u>

- Raspberry Pi
- Small camera
- Buttons
- Small commercial programmable robot (Optional)
- Small screen
- LEDs

<u>Facilities</u>

None

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